

ITA0606 – MACHINE LEARNING

EXP 5

PROGRAM:

Implementation of K-Nearest Neighbours (K-NN) in Python

Import required libraries

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score, confusion_matrix

Step 1: Load the Iris dataset

iris = load_iris()

X = iris.data

y = iris.target

Step 2: Split the dataset into training and testing data

X_train, X_test, y_train, y_test = train_test_split(

 X, y,

 test_size=0.3,

 random_state=42

)

Step 3: Create K-NN classifier

k = 3 means the 3 nearest neighbours are considered

model = KNeighborsClassifier(n_neighbors=3)

```
# Step 4: Train the model
```

```
model.fit(X_train, y_train)
```

```
# Step 5: Predict the test data
```

```
y_pred = model.predict(X_test)
```

```
# Step 6: Display actual and predicted values
```

```
print("Actual Classes:")
```

```
print(y_test)
```

```
print("\nPredicted Classes:")
```

```
print(y_pred)
```

```
# Step 7: Calculate accuracy
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("\nAccuracy:")
```

```
print(accuracy * 100, "%")
```

```
# Step 8: Display confusion matrix
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
print("\nConfusion Matrix:")
```

```
print(cm)
```

```
# Step 9: Display class names
```

```
print("\nClass Names:")
```

```
print(iris.target_names)
```

```

# Step 10: Predict a new sample
new_flower = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(new_flower)

print("\nNew Flower:")
print(new_flower)

print("\nPredicted Class:")
print(iris.target_names[prediction[0]])

```

OUTPUT:

```

... Actual Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Predicted Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Accuracy:
100.0 %

Confusion Matrix:
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]

Class Names:
['setosa' 'versicolor' 'virginica']

New Flower:
[[5.1, 3.5, 1.4, 0.2]]

Predicted Class:
setosa

```